ASSET LIFECYCLE MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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in partial fulfilment of the End Semester Examination for the course

CB23F35 – LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

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Computer Science and Engineering



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2026 – 2027

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BONAFIDE CERTIFICATE

Certified that this project report titled "Asset Lifecycle Management System" is the bonafide work of SANDEEP K (2116240701464), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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DECLARATION

I, SANDEEP K (2116240701464), hereby declare that the project report titled "Asset Lifecycle Management System", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report, including its data model, forms, reports, pages, Deluge scripts, workflows, security permission profiles, and scheduled automations, was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature, code snippet, or external platform resource used in this work has been duly acknowledged in the report.

Place:
Date:

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SANDEEP K

ABSTRACT

Organisations of every size depend on a large and varied base of physical and digital assets, such as laptops, desktops, servers, projectors, networking equipment, software licences and furniture. The value of these assets is realised only when they are procured, assigned, maintained, verified and finally retired in a controlled manner. In many institutions this lifecycle is still managed through disconnected spreadsheets, e-mail threads and paper files. This produces fragmented records, unclear accountability for who holds which asset, missed warranty windows, informal handling of faults, unverified inventories and poorly documented disposals.

This project presents the Asset Lifecycle Management System, a cloud application designed and deployed on the Zoho Creator low-code platform. The system comprises eleven interrelated forms (Vendor Supplier, Asset Purchase, Purchase Items, Asset Registration, Warranty, Maintenance Request, Asset Transfer, Asset Audit, Asset Retirement Disposal, Department and Employee User) joined by lookup fields. Deluge scripting notifies an employee when an asset is assigned to them, prevents a critical maintenance request from being saved without a technician, stamps repair completion dates automatically and initialises every transfer request as pending. A daily scheduled workflow recalculates each warranty as Active, Expiring Soon (within 30 days) or Expired, and a 14-stage blueprint stage model describes the lifecycle from purchase initiation to asset disposal.

The user interface features a ZML dashboard with spend, tax, unit-cost and renewal-cost indicators, warranty status counters, two scatter analytics charts and quick links, supported by twenty list and Kanban reports with aggregates and conditional formatting. Role-Based Access Control is implemented through the built-in Administrator profile and six custom profiles (Asset Administrator, Procurement Officer, IT Maintenance Technician, Department Head, Vendor Representative and Employee User), together with a CEO role in the hierarchy. The result is a single, auditable platform that improves asset visibility, accountability and cost control across the complete lifecycle.

Keywords: Low-Code Application Development; Zoho Creator; Deluge Scripting; IT Asset Management; Asset Lifecycle; Warranty Monitoring; Workflow Automation; Role-Based Access Control (RBAC).

PROJECT AT A GLANCE

Attribute	Details
Project Title	Asset Lifecycle Management System
Application Domain	Enterprise and Institutional IT Asset Management (Procurement to Disposal)
Student Name & Register No.	SANDEEP K · 2116240701464
Zoho Creator Edition	Enterprise / Academic Cloud Edition
Application Name (Link Name)	ASSET LIFECYCLE MANAGEMENT SYSTEM (asset-lifecycle-management-system)
Published / Live URL	https://creatorapp.zoho.in/mohanprasath.p.2024.cse_r285/asset-lifecycle-management-system
Demo Credentials (per role)	Administrator, Asset Administrator, Procurement Officer, IT Maintenance Technician, Department Head, Employee User, Vendor Representative
Number of Forms / Subforms	11 Forms / 0 Subforms (vendor_supplier, asset_purchase, purchase_items, asset_registration, warranty, maintenance_request, asset_transfer, asset_audit, asset_retirement_disposal, department, employee_user)
Number of Reports (by type)	List: 11 ▏ Kanban: 9 ▏ Summary Matrix: 0 ▏ Total: 20 Reports
Number of Pages / Dashboards	1 Page (DashBoard) with 4 KPI cards, 2 scatter charts, warranty status counters, quick links and an embedded report
Number of Workflows	Form: 4 ▏ Schedule: 1 ▏ Blueprint: 1 (14 stages defined on Asset Purchase) ▏ Batch: 0
Deluge Functions / Actions	5 custom Deluge scripts (4 form-event scripts and 1 daily scheduled script)
Integrations	Built-in Deluge sendmail notification (asset-assignment e-mail to the employee)
Roles & Profiles Defined	Administrator, Asset Administrator, Procurement Officer, IT Maintenance Technician, Department Head, Vendor Representative (portal), Employee User (portal); role hierarchy: CEO
Environments Used	Development and Production Cloud Workspace
Date of Production Deployment	09-November-2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AMC	Annual Maintenance Contract
API	Application Programming Interface
CO / PO	Course Outcome / Program Outcome
Deluge	Data Enriched Language for the Universal Grid Environment
ER	Entity–Relationship
FR	Functional Requirement
GST	Goods and Services Tax
HOD	Head of the Department
ISO	International Organization for Standardization
IT	Information Technology
ITAM	IT Asset Management
KPI	Key Performance Indicator
NFR	Non-Functional Requirement
RBAC	Role-Based Access Control
TLS	Transport Layer Security
UAT	User Acceptance Testing
UI / UX	User Interface / User Experience
USD	United States Dollar (currency field type)
ZML	Zoho Markup Language

CHAPTER 1
INTRODUCTION

1.1 Background

Colleges, hospitals, offices and factories invest heavily in physical and digital assets such as laptops, desktops, servers, projectors, printers, networking equipment, furniture and software licences. Every asset passes through a lifecycle: planning and purchase, receipt and registration, assignment to a person or department, day-to-day use, maintenance and warranty cover, transfer between users or locations, periodic physical verification, and finally retirement and disposal. International guidance such as ISO 55000 for asset management and ISO/IEC 19770-1 for IT asset management stresses that value, risk and cost can be controlled only when information about the whole lifecycle is complete, current and trusted.

In practice, many mid-sized institutions track this lifecycle in shared spreadsheets, e-mail conversations and paper folders. Purchase records, invoices, warranty documents, repair history and audit notes live in different places and are maintained by different people. The Asset Lifecycle Management System, engineered on Zoho Creator, brings every stage of the lifecycle into one secure cloud application with relational data, automated business rules and role-specific access.

1.2 Problem Statement

Manual or spreadsheet-based asset tracking introduces several recurring failure points:

- Fragmented Records: Purchase, registration, warranty and maintenance information is scattered across files, so no single source of truth exists for an asset.
- Weak Accountability: There is no dependable record of who holds an asset, where it is located, or when its owner, department or location last changed.
- Missed Warranty Windows: Warranty expiry is noticed only after a failure, which leads to avoidable repair and replacement costs.
- Unstructured Maintenance Handling: Faults are reported informally, critical issues are not guaranteed a technician, and repair cost and completion date go unrecorded.

- Unverified Inventory: Without a structured audit, it is unclear whether an asset was physically found, whether its serial number, assignee and location were verified, and in what condition it is.
- Opaque Retirement and Spend: Disposal reasons, methods, book value and recovered value are rarely recorded, and management has no consolidated view of spend, tax and renewal cost.

1.3 Objectives

The primary objectives of this project are:

- To design a relational data model covering the full asset lifecycle: vendors, purchases, purchase items, registered assets, warranties, maintenance requests, transfers, audits, retirement and disposal, departments and employees.
- To implement reactive Deluge business logic that notifies assignees, validates critical maintenance requests, stamps repair completion dates and initialises transfer approvals.
- To automate warranty governance through a daily scheduled workflow that classifies every warranty as Active, Expiring Soon or Expired.
- To design a ZML dashboard and a set of list and Kanban reports with aggregates and conditional formatting that give real-time visibility of cost, status and condition.
- To establish Role-Based Access Control so that procurement, asset administration, IT support, department heads, vendors and employees each see only the modules relevant to their work.

1.4 Scope of the Project

In Scope: Master data for departments, employees and vendors; purchase orders with line items, tax and payment tracking; asset registration with automatic Asset Id and status; warranty records with daily status monitoring; maintenance requests with priority, technician, cost and completion tracking; asset transfer requests with approval status; structured asset audits; retirement and disposal records with book value and disposal value; a dashboard; 20 list and Kanban reports; and profile-based access control.

Out of Scope: Integration with ERP or accounting packages, barcode or RFID scanning hardware, automatic depreciation calculation, and multi-currency or multi-site consolidation.

1.5 Justification for a Low-Code Approach

A conventional full-stack build would need a database schema, back-end services, a responsive front end, authentication, scheduling and hosting to be created and maintained separately. Asset policies, categories and approval rules change frequently, so hard-coded solutions become expensive to modify. Zoho Creator provides relational forms, drag-and-drop report and page builders, managed cloud hosting, a profile-based permission model and the Deluge scripting language for custom logic and scheduled automation. This allows the whole lifecycle application to be built, tested and adapted in weeks rather than months.

1.6 Organisation of the Report

Chapter 2 reviews the existing manual process and related literature. Chapter 3 specifies functional and non-functional requirements. Chapter 4 presents the architecture, data model and access design. Chapter 5 details the implementation, including forms, reports, the dashboard and Deluge scripts. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, security and governance. Chapter 8 evaluates the results through an operational walkthrough. Chapter 9 concludes with future enhancements.

CHAPTER 2

LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In a typical institution, a department requests equipment by e-mail or a paper note. After approval and delivery, stores staff enter the item into a shared spreadsheet register. Invoices and warranty cards are filed in cupboards or saved in personal folders. When an asset moves to another person or room, the change is communicated by phone or message and the register is updated later, if at all. Faults are reported verbally to the IT desk, and a yearly physical verification is carried out against a printed list. There is no audit trail of who changed what and when.

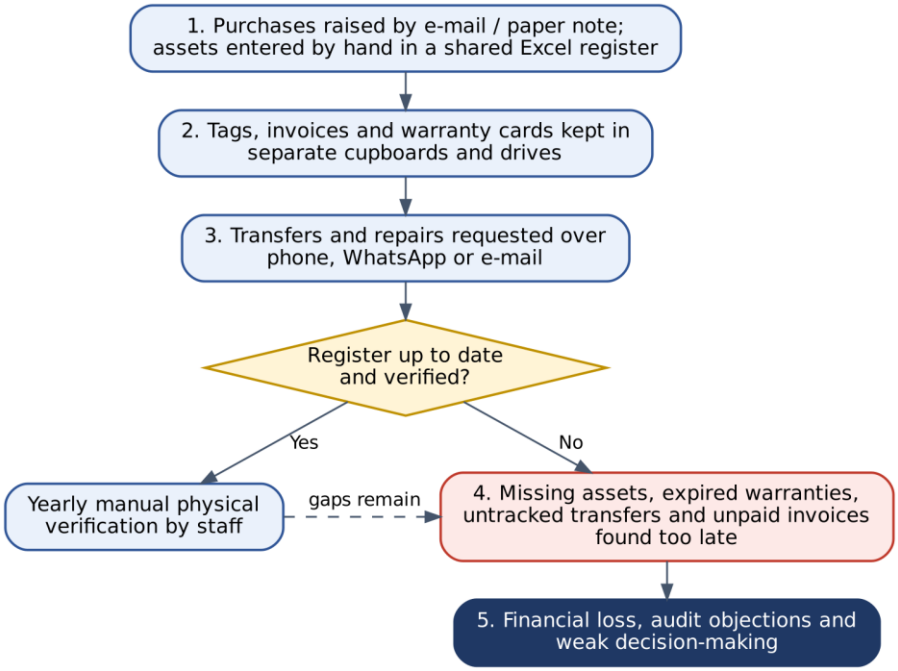


Figure 2.1: As-Is Process Flow of the Existing Manual Asset Management Process

2.2 Limitations of the Existing System

- Data Inconsistency: Duplicate and conflicting spreadsheet versions prevent a single source of truth for each asset.
- Absence of Guardrails: A critical fault can be logged with nobody assigned to it, and an asset can be transferred without any recorded approval.
- Missed Warranty Cover: Expiry dates are held in documents rather than monitored, so cover lapses unnoticed.

- Poor Traceability: Ownership, location and condition histories are not captured in a structured manner.
- No Consolidated Visibility: Total spend, tax paid, renewal cost and book value must be compiled manually from several files.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development Cycle	Several months (separate front end, API and database build)	A few weeks using visual modelling and Deluge
Database Architecture	Manual SQL schema design, migrations and foreign keys	Visual forms with lookup fields
Access Control (RBAC)	Custom authentication and permission tables	Built-in permission profiles and roles
Hosting & Maintenance	Dedicated servers, SSL certificates and deployment pipelines	Managed cloud infrastructure
Workflow Automation	Separate cron jobs or scheduler services	Form events and scheduled Deluge workflows
Cost & Maintenance	High developer effort and ongoing server cost	Platform subscription with low maintenance effort

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) & Year	Title / Source	Key Findings	Relevance to this Project
1	A. C. Bock and U. Frank, 2021	Low-Code Platform	Describes low-code platforms as model-driven environments that reduce hand-written boilerplate.	Underpins the use of visual forms and lookups.
2	A. Sahay et al., 2020	Supporting the Understanding and Comparison of Low-Code Development Platforms	Compares architectural capabilities of commercial low-code platforms.	Supports the choice of Zoho Creator.
3	Y. Luo et al., 2021	Characteristics and Challenges of Low-Code Development	Reports agility benefits and platform-boundary constraints from practitioners.	Guides validation rules and limits handled in Deluge.

S.No.	Author(s) & Year	Title / Source	Key Findings	Relevance to this Project
4	ISO, 2014	ISO 55000: Asset management – Overview, principles and terminology	Sets out principles for realising value from assets through lifecycle management.	Basis for modelling purchase-to-disposal stages.
5	ISO/IEC, 2017	ISO/IEC 19770-1: IT asset management systems – Requirements	Specifies processes for managing IT assets across their lifecycle.	Shapes the forms for registration, audit and retirement.
6	Zoho Corporation, 2026	Zoho Creator Help Documentation	Technical reference for forms, reports, pages and permissions.	Reference for schema, reports and profiles.
7	Zoho Corporation, 2026	Deluge Script Reference Guide	Syntax reference for fetch, loops, sendmail and form events.	Direct guide for the five scripts.

2.5 Proposed System Overview

The proposed Asset Lifecycle Management System replaces spreadsheets and paper files with one synchronised cloud application. Eleven related forms capture every lifecycle event, Deluge rules enforce data quality and send notifications, a daily schedule keeps warranty status current, a dashboard and 20 reports give management live visibility, and permission profiles ensure each user works only within their responsibilities.

CHAPTER 3

REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role / Profile	Description	Key Responsibilities in Application
Administrator	System owner with all permissions.	Configure the application, manage users and profiles, and monitor overall asset health.
Asset Administrator	Custodian of the asset register.	Register new assets and manage the lifecycle from registration to retirement, including transfers, audits and disposals.
Procurement Officer	Purchasing and vendor coordinator.	Create and manage vendors, purchase orders and purchase items, and track payment status.
IT Maintenance Technician	Support engineer.	Receive maintenance requests, work them by priority and issue type, and record cost and completion.
Department Head	Head of an academic or business unit.	Request asset transfers for team members, approve transfers and view audit results.
Vendor Representative (Portal)	External supplier contact.	Maintain vendor contact details and warranty or service-contract information.
Employee User (Portal)	Any employee who uses an asset.	Report hardware or software issues through maintenance requests and view transfer and audit information.
CEO (Role)	Top of the role hierarchy.	Access the data of all subordinate users for oversight.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Feature / Requirement	Role(s)	Priority	Detailed Description
FR-01	Master Data Management	Administrator, Department Head	High	The system shall maintain departments (name, head, location, contact e-mail, status) and employees (name, e-mail, phone, department, designation, joining date, employment status).
FR-02	Vendor Management	Procurement Officer, Vendor Representative	High	The system shall store vendors with an auto-generated Vendor Id, contact person, GST number, category and status.

FR ID	Feature / Requirement	Role(s)	Priority	Detailed Description
FR-03	Purchase Recording	Procurement Officer	High	The system shall record purchase orders with vendor, invoice, quantity, unit cost, tax, subtotal, total, payment and approval status, and attach line items through Purchase Items.
FR-04	Asset Registration	Asset Administrator	High	The system shall register assets with an auto-generated Asset Id, category, serial number, cost, assignee, location and a default status of In Stock.
FR-05	Assignment Notification	System / Deluge	Medium	When an asset is registered, the system shall e-mail the assigned employee with the asset name and category.
FR-06	Warranty Management	Vendor Representative, Administrator	High	The system shall maintain warranty type, period, coverage, document, renewal requirement and renewal cost for each asset.
FR-07	Automated Warranty Status	Scheduled Workflow	High	A daily workflow shall set status to Expired, Expiring Soon (within 30 days) or Active, leaving Renewed warranties unchanged.
FR-08	Maintenance Handling	Employee, IT Technician	High	The system shall capture issue type, priority, technician, estimated and actual cost and repair status; a Critical request must have a technician; completion date is set automatically.
FR-09	Asset Transfer	Department Head, Asset Administrator	Medium	The system shall record previous and new employee, department and location with a reason, and initialise approval status as Pending.
FR-10	Asset Audit	Asset Administrator	Medium	The system shall record whether an asset was found and its condition, and whether serial number, assignee and location were verified, with a result and status.
FR-11	Retirement and Disposal	Asset Administrator	Medium	The system shall record disposal reason, method, original cost, book value, disposal value, approval status and certificate reference.
FR-12	Dashboard and Reports	All Roles (by permission)	Medium	The system shall present KPIs, warranty status counters, charts and 20 list and Kanban reports with totals and colour coding.

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Specification
NFR-01	Performance	List reports, Kanban boards and the dashboard should load within about two seconds for a typical institutional dataset.
NFR-02	Security & RBAC	Each profile can open only the modules and reports assigned to it; portal users see a restricted set of modules.
NFR-03	Data Integrity	Asset references use lookup fields; validation rules block incomplete critical maintenance requests.
NFR-04	Availability	The application is hosted on Zoho Creator's managed cloud infrastructure and is accessible through any modern browser.
NFR-05	Usability	Responsive layouts for web, phone and tablet with simple forms, clear field labels and informative validation messages.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Cloud Development Platform	Zoho Creator (Enterprise Low-Code Platform)
Workflow Scripting Language	Deluge (Data Enriched Language for the Universal Grid Environment)
User Interface Technology	Zoho UI Builder and Zoho Markup Language (ZML)
Web Browser Support	Google Chrome, Microsoft Edge, Mozilla Firefox (latest versions)
Client Hardware	Intel Core i3/i5 or higher, 8 GB RAM, active Internet connection
Notification Subsystem	Zoho Creator native sendmail communication engine

3.5 Use Case Model

The interactions between the actors (Administrator, Asset Administrator, Procurement Officer, IT Maintenance Technician, Department Head, Employee User, Vendor Representative and the CEO role) and the system are modelled in Figure 3.1.

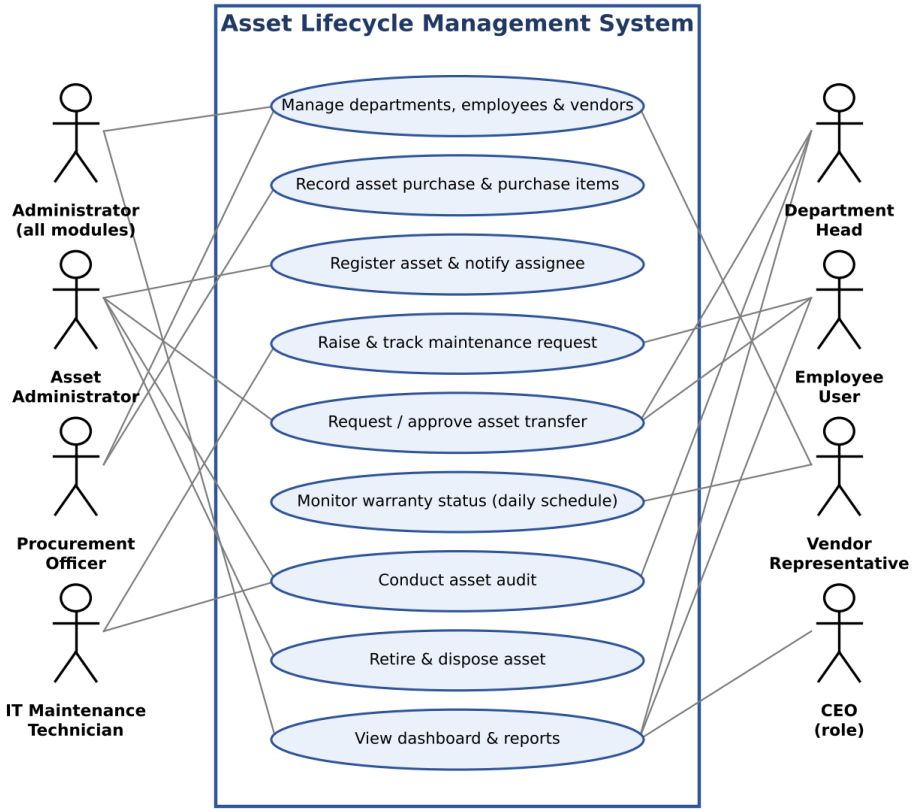


Figure 3.1: Use Case Diagram of the Asset Lifecycle Management System

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

The application follows a layered architecture made up of the User Presentation Layer, the Application Layer, the Data and Logic Layer and the Automated Communication Layer, as shown in Figure 4.1.

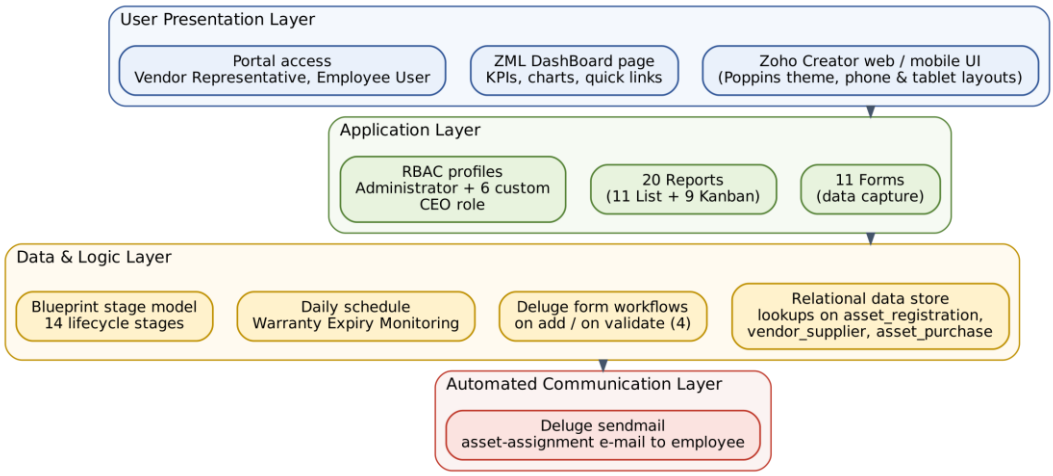


Figure 4.1: System Architecture of the Asset Lifecycle Management System

4.2 Data Model & Relational Schema

The data model consists of eleven forms, grouped by purpose:

- Master data: department (name, head, location, contact e-mail, status), employee_user (name, contact, department, designation, employment status) and vendor_supplier (auto Vendor Id, contact, GST number, category, status).
- Procurement: asset_purchase (purchase order, invoice, quantity, unit cost, tax, total, payment and approval status) with child line items in purchase_items, linked through the parent_purchase lookup.
- Core register: asset_registration (auto Asset Id, category, brand, model, serial number, cost, assignee, location, status and warranty summary).
- Lifecycle events: warranty, maintenance_request, asset_transfer, asset_audit and asset_retirement_disposal, each linked to asset_registration through an asset lookup.

Seven lookup fields enforce referential links: asset_purchase.vendor → vendor_supplier; purchase_items.parent_purchase → asset_purchase; and the asset field of warranty, maintenance_request, asset_transfer, asset_audit and asset_retirement_disposal → asset_registration. In asset_registration, the vendor, department and assigned employee are stored as text for fast data entry; these are shown as dashed (logical) associations in Figure 4.2.

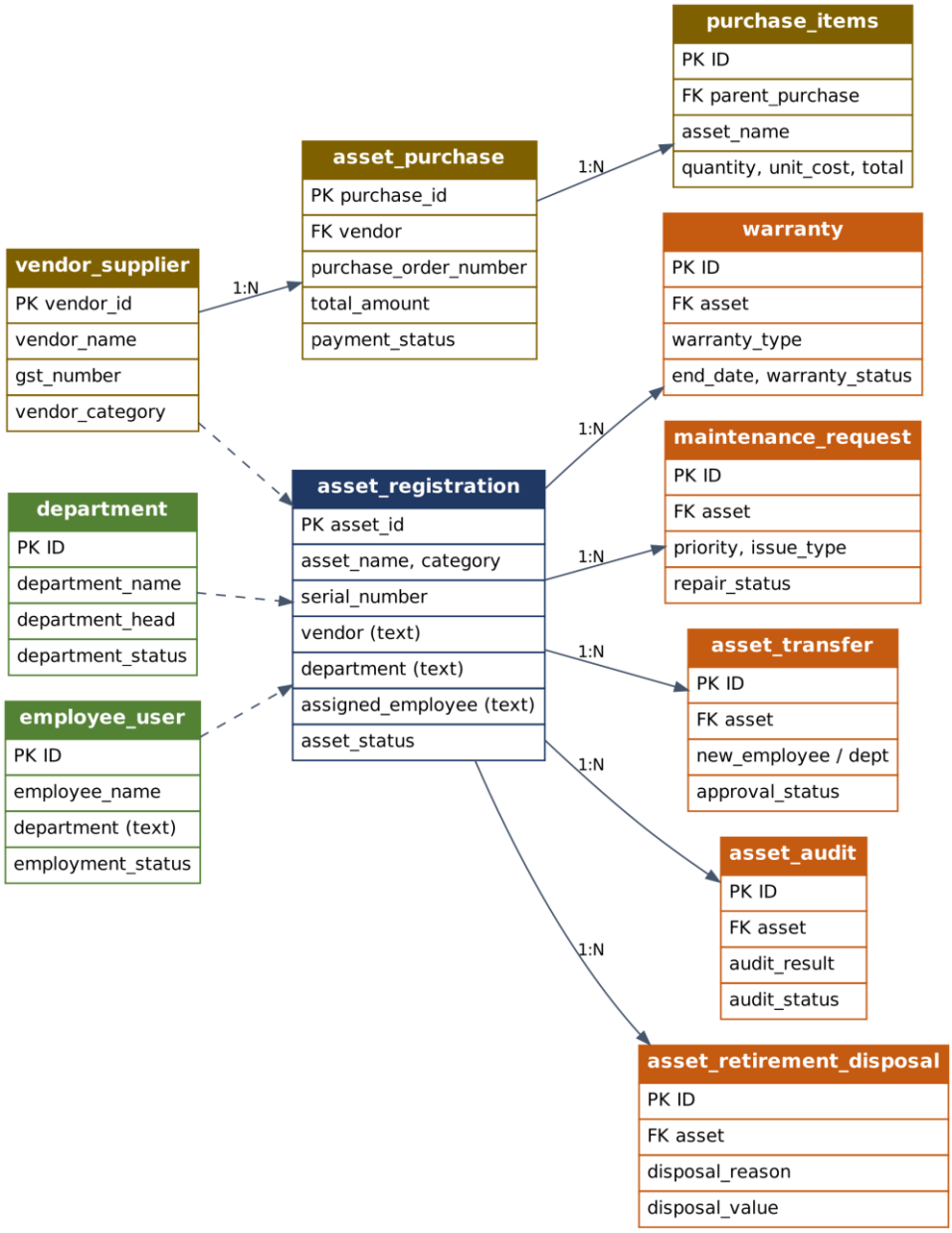


Figure 4.2: Entity–Relationship (ER) Diagram of the Application

4.3 Form Field Specifications

The eleven forms contain 128 data fields in total and use 13 distinct field types: Auto Number, Text, Picklist, Lookup, Date, Date-Time, Currency (USD), Decimal, Number,

Textarea, Radio Buttons, Email and Phone Number, in addition to the composite Name field. Tables 4.1 to 4.3 describe the three principal forms.

Table 4.1: Field Specifications – Asset Registration Form

Field Label	Link Name	Field Type	Description / Validation
Asset Id	asset_id	Auto Number	Unique identifier generated automatically, starting at 1.
Asset Name	asset_name	Text	Name of the asset.
Asset Category	asset_category	Picklist	Projector, Laptop, Software License, Monitor, Desktop, Server, Furniture, Printer, Mobile Device, Networking Equipment, Other.
Brand / Model	brand, model	Text	Manufacturer and model of the asset.
Serial Number	serial_number	Text	Manufacturer serial number, checked during audits.
Purchase Date	purchase_date	Date	Date of purchase.
Purchase Cost	purchase_cost	Currency (USD)	Acquisition cost; totalled in reports.
Vendor	vendor	Text	Supplier of the asset.
Department	department	Text	Owning department.
Assigned Employee	assigned_employee	Text	Employee holding the asset; matched with the employee's first name by the notification workflow.
Current Location	current_location	Text	Room or site where the asset is kept.
Asset Status	asset_status	Picklist	Assigned, Lost, In Use, Retired, Under Maintenance, In Stock, Damaged, Disposed; initial value In Stock.
Warranty Start / End Date	warranty_start_date, warranty_end_date	Date	Warranty validity period.
Warranty Provider	warranty_provider	Text	Organisation providing warranty cover.
Invoice	invoice	Text	Invoice reference.
Remarks	remarks	Textarea	Free-text notes.

Table 4.2: Field Specifications – Asset Purchase Form

Field Label	Link Name	Field Type	Description / Validation
Purchase Id	purchase_id	Auto Number	Unique purchase reference generated automatically.
Purchase Date	purchase_date	Date	Date the order was placed.
Vendor	vendor	Lookup (vendor_supplier)	Supplier chosen from the vendor master; new vendors can be added inline.

Field Label	Link Name	Field Type	Description / Validation
Purchase Order Number	purchase_order_number	Text	Purchase order reference.
Invoice Number	invoice_number	Text	Supplier invoice reference.
Asset Category	asset_category	Picklist	Same eleven categories as the asset register.
Quantity	quantity	Number	Number of units purchased.
Unit Cost	unit_cost	Currency (USD)	Cost per unit; totalled in reports and on the dashboard.
Tax Percentage / Tax Amount	tax_percentage, tax_amount	Decimal / Currency	Tax rate and the resulting tax value.
Subtotal / Total Amount	subtotal, total_amount	Currency (USD)	Pre-tax and final values of the order.
Payment Status	payment_status	Picklist	Paid, Pending, Partially Paid.
Payment Date	payment_date	Date	Date payment was made.
Invoice File	invoice_file	Text	Reference to the invoice document.
Purchase Approval	purchase_approval	Picklist	Approved, Rejected, Pending.
Remarks	remarks	Textarea	Free-text notes.

Table 4.3: Field Specifications – Maintenance Request Form

Field Label	Link Name	Field Type	Description / Validation
Asset	asset	Lookup (asset_registration)	Asset needing service; displayed by Asset Id.
Reported By	reported_by	Text	Person who raised the request.
Request Date	request_date	Date-Time	When the issue was reported.
Issue Type	issue_type	Picklist	Software Problem, Hardware Failure, Preventive Maintenance, Physical Damage, Network Problem, Other.
Problem Description	problem_description	Textarea	Details of the fault.
Priority	priority	Picklist	Critical, High, Medium, Low; a Critical request requires a technician.
Assigned Technician	assigned_technician	Text	Technician responsible for the repair.
Estimated / Actual Cost	estimated_cost, actual_cost	Currency (USD)	Quoted and final repair cost.
Repair Status	repair_status	Picklist	New, Assigned, In Progress, Waiting for Parts, Completed, Cancelled.
Completion Date	completion_date	Date	Set automatically when status becomes Completed.

Field Label	Link Name	Field Type	Description / Validation
Service Report	service_report	Text	Reference to the service report.
Remarks	remarks	Textarea	Free-text notes.

The remaining forms follow the same conventions. Asset Audit records Asset Found, Serial Number Verified, Assigned Person Verified and Location Verified as Yes/No radio buttons, together with Asset Condition (Excellent, Fair, Good, Non-Functional, Damaged), Audit Result (Missing, Verified, Discrepancy Found, Damaged) and Audit Status (Open, Under Investigation, Resolved, Closed). Warranty uses Warranty Type (AMC, Extended, Standard, Comprehensive) and Warranty Status (Active, Expired, Renewed, Expiring Soon).

4.4 Process and Workflow Design

An asset moves through the lifecycle shown in Figure 4.3. The Asset Purchase form carries a blueprint stage model of 14 stages: Asset Purchase Initiated, Purchase Approval Pending, Purchase Rejected, Asset Acquired, Asset Retired, Asset In Transit, Asset Received, Asset Damaged on Arrival, Asset Lost, Asset Active, Asset Returned to Vendor, Asset Maintenance Required, Asset Transferred and Asset Disposed. In operation, the registered asset's Asset Status (In Stock, Assigned, In Use, Under Maintenance, Damaged, Lost, Retired, Disposed) tracks the current position of each asset.

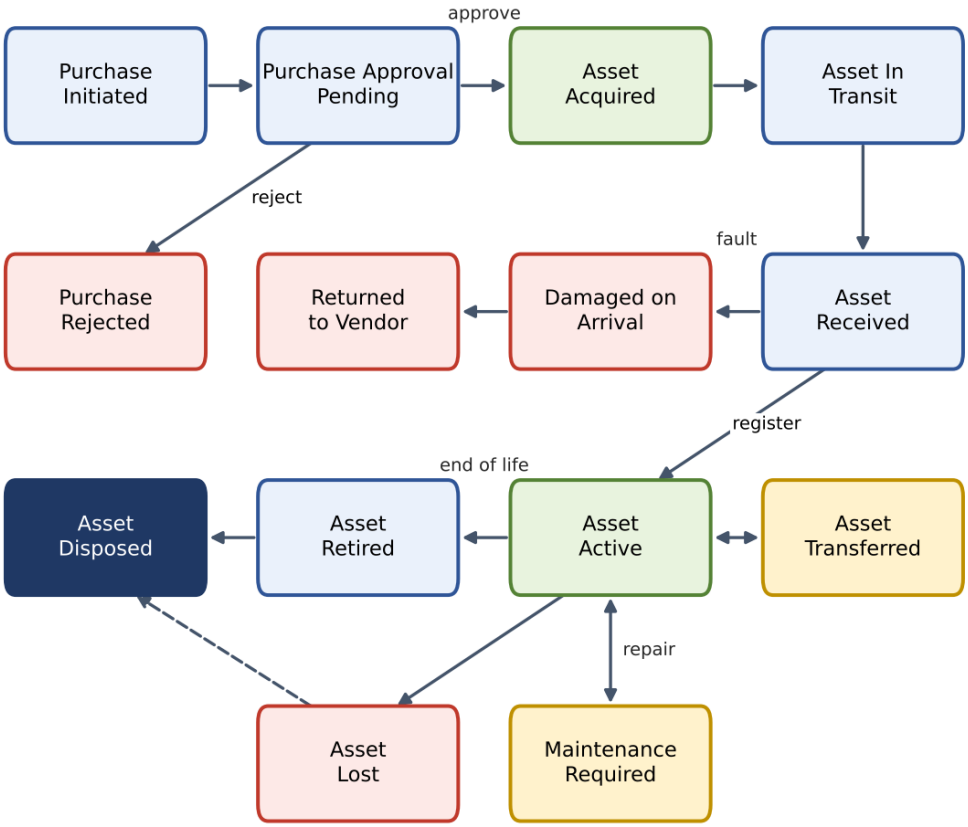


Figure 4.3: End-to-End Asset Lifecycle and State Transition Diagram

4.5 User Interface and Navigation Design

The left navigation lists the Dashboard followed by one section for each form and its reports: Warranty, Asset Purchase, Maintenance Request, Purchase Items, Asset Retirement Disposal, Employee User, Asset Transfer, Department, Asset Registration, Vendor Supplier and Asset Audit. Approvals and shared Reports sections are provided by the platform. The application uses theme 11 with the Poppins font, a two-column layout on phones and a sliding-pane layout on tablets.

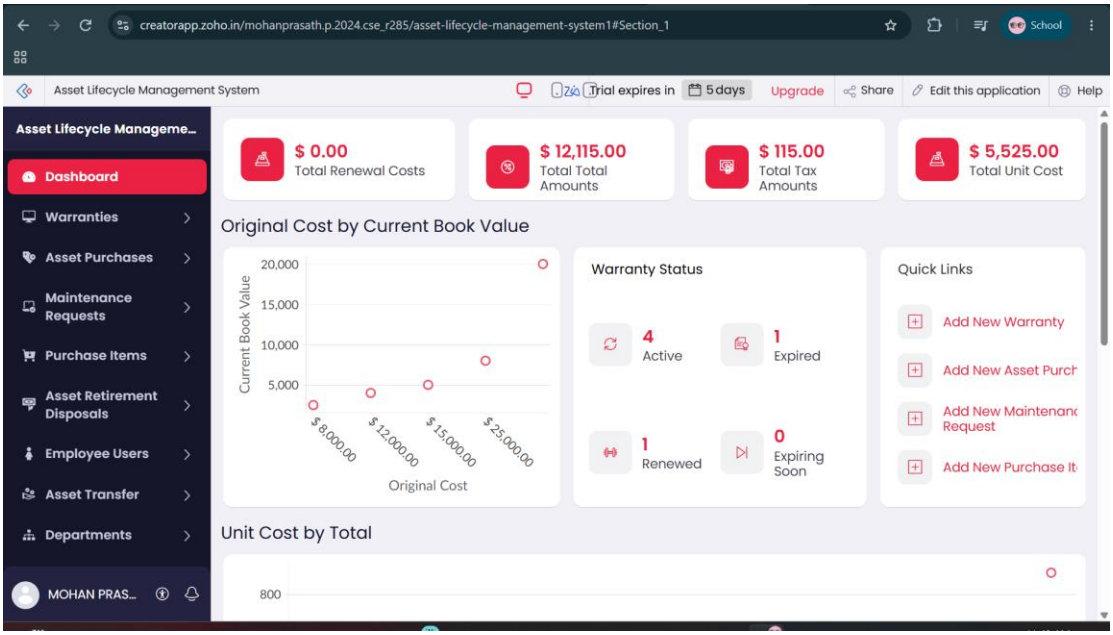


Figure 4.4: Navigation Structure and Wireframe of the Dashboard

4.6 Role-Based Access Control Design

Table 4.4: Role-Based Access Control Permission Matrix

Module	Admin-istrator	Asset Admin.	Procure-ment Officer	IT Techni-cian	Dept. Head	Vendor Rep. (Portal)	Employee (Portal)
asset_registration	Full	Full	—	—	—	—	—
asset_purchase	Full	—	Full	—	—	—	—
purchase_items	Full	—	Full	—	—	—	—
vendor_supplier	Full	—	Full	—	—	Full	—
warranty	Full	—	—	—	—	Full	—
maintenance_request	Full	—	—	Full	—	—	Full
asset_transfer	Full	Full	—	—	Full	—	View
asset_audit	Full	Full	—	View	View	—	View
asset_retirement_disposal	Full	Full	—	—	—	—	—
department	Full	—	—	—	View, Modify	—	—
employee_user	Full	—	—	—	—	—	—

"Full" means Create, View All, Modify All, Import, Export and tab access, together with view, edit and delete rights on the module's reports. "View" means View All with view-only report access. The Administrator profile carries all permissions. A CEO role is defined at the top of the role hierarchy so that its holder can access the data of all subordinate users.

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application Asset Lifecycle Management System was created in the Zoho Creator developer workspace with the date format dd-MMM-yyyy and a 24-hour time format. It provides 11 forms, 20 reports, 1 dashboard page, 4 form workflows, 1 scheduled workflow and a 14-stage blueprint stage model, arranged in a navigation menu of 12 sections plus the system sections for approvals, preferences and shared reports.

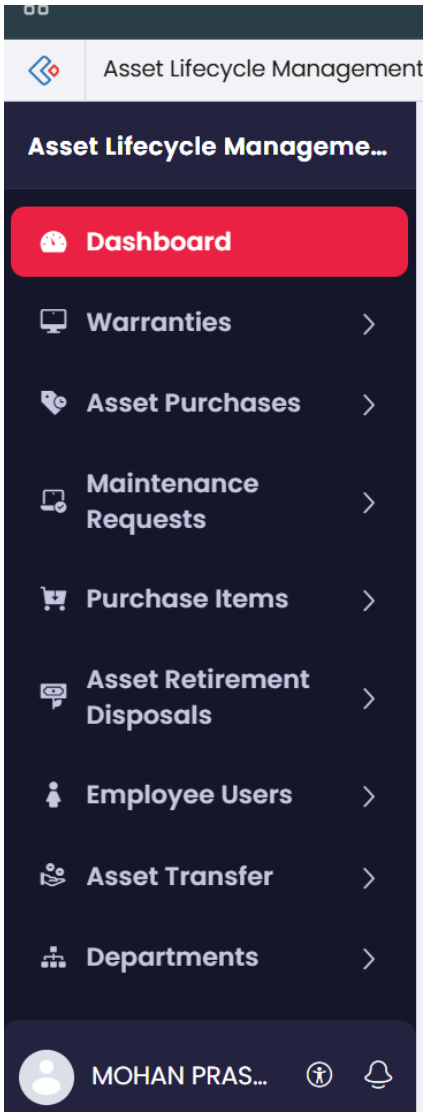


Figure 5.1: Application Home and Navigation Sections

5.2 Forms and Data Capture

Table 5.1: Forms Implemented in the Application

Form (Link Name)	Fields	Purpose
Vendor Supplier (vendor_supplier)	9	Vendor master with auto Vendor Id, contact details, GST number, category and status.
Asset Purchase (asset_purchase)	17	Purchase order header with vendor lookup, tax, totals, payment and approval; carries the blueprint stage set.
Purchase Items (purchase_items)	9	Line items of a purchase (asset, brand, model, quantity, unit cost, warranty period, total) linked by parent_purchase.
Asset Registration (asset_registration)	18	Central asset register with auto Asset Id, status, assignment and warranty summary.
Warranty (warranty)	11	Warranty type, validity, coverage, document, renewal requirement and cost.
Maintenance Request (maintenance_request)	13	Fault reports with priority, technician, costs, repair status and completion date.
Asset Transfer (asset_transfer)	13	Change of employee, department or location with reason and approval status.
Asset Audit (asset_audit)	13	Physical verification of presence, condition, serial number, assignee and location.
Asset Retirement Disposal (asset_retirement_disposal)	12	Retirement reason, method, cost, book value, disposal value and certificate.
Department (department)	5	Department master with head, location, contact e-mail and status.
Employee User (employee_user)	8	Employee directory with name, contact, department, designation and employment status.

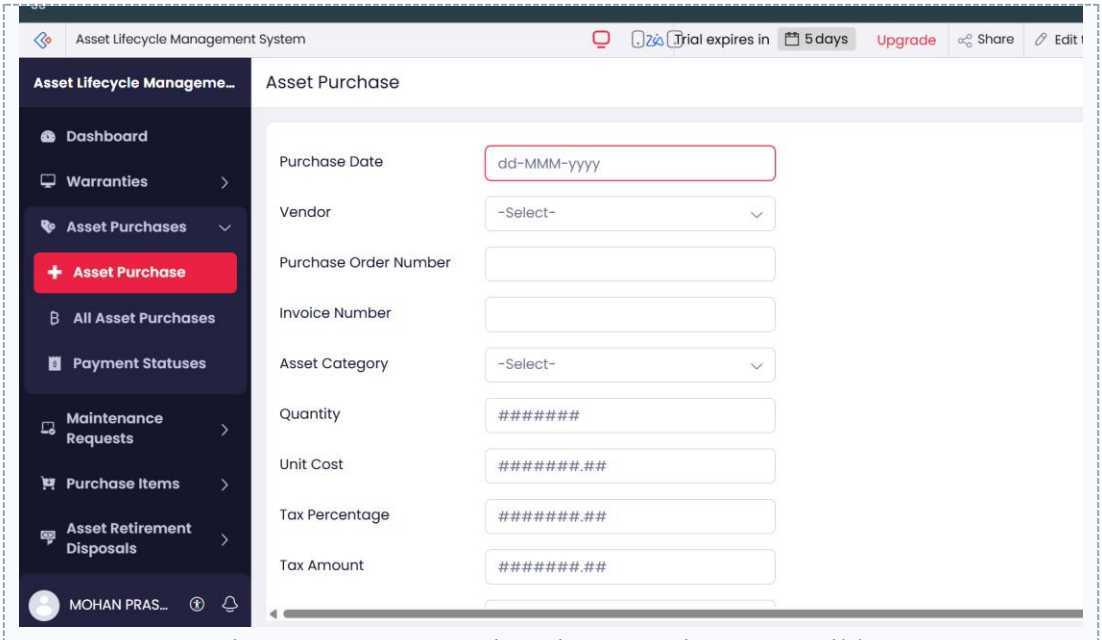


Figure 5.2: Asset Registration Form in Form Builder

5.3 Reports and Analytics Views

Twenty reports were configured: eleven list reports and nine Kanban boards. Eight list reports carry 26 conditional-formatting rules that colour-code key status fields, and list and Kanban reports display total, average, minimum and maximum for currency fields.

Table 5.2: Reports Implemented in the Application

Report Name	Report Type	Source Form	Aggregates / Formatting / Grouping
All Warranties	List	warranty	Aggregates on Renewal Cost; colour coding on Warranty Type (AMC, Comprehensive, Extended).
Warranty Types	Kanban	warranty	Cards grouped by Warranty Type.
All Asset Purchases	List	asset_purchase	Aggregates on Unit Cost, Tax Amount, Subtotal, Total Amount; colour coding on Payment Status.
Payment Statuses	Kanban	asset_purchase	Grouped by Payment Status (Paid, Pending, Partially Paid).
All Maintenance Requests	List	maintenance_request	Aggregates on Estimated and Actual Cost; colour coding on Priority (Critical, High, Medium, Low).
Issue Types	Kanban	maintenance_request	Grouped by Issue Type.
All Purchase Items	List	purchase_items	Aggregates on Unit Cost and Total.
All Asset Retirement Disposals	List	asset_retirement_disposal	Aggregates on Original Cost, Book Value, Disposal Value; colour coding on Asset Condition.

Report Name	Report Type	Source Form	Aggregates / Formatting / Grouping
Disposal Reasons	Kanban	asset_retirement_disposal	Grouped by Disposal Reason.
All Employee Users	List	employee_user	Colour coding on Employment Status (Active, On Leave, Resigned, Terminated).
Employment Statuses	Kanban	employee_user	Grouped by Employment Status.
Asset Transfer Report	List	asset_transfer	Colour coding on Approval Status (Approved, Pending, Rejected).
Reasons	Kanban	asset_transfer	Grouped by Transfer Reason.
All Departments	List	department	Department master list.
All Asset Registrations	List	asset_registration	Aggregate on Purchase Cost.
Asset Statuses	Kanban	asset_registration	Grouped by Asset Status.
All Vendor Suppliers	List	vendor_supplier	Colour coding on Vendor Category (Networking, Office Equipment, Other).
Vendor Categories	Kanban	vendor_supplier	Grouped by Vendor Category.
All Asset Audits	List	asset_audit	Colour coding on Asset Condition (Excellent, Fair, Non-Functional).
Asset Conditions	Kanban	asset_audit	Grouped by Asset Condition.

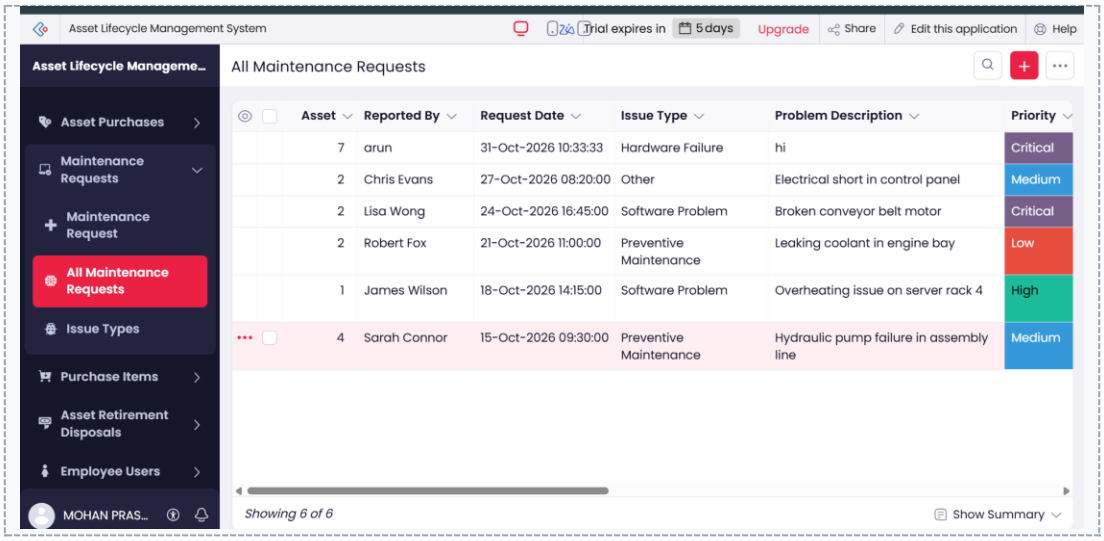


Figure 5.4: Maintenance Request Report with Priority Colour Coding

5.4 Pages and Interactive Dashboards

The DashBoard page is built in ZML. Its first row holds four KPI cards that sum Renewal Cost from Warranty and Total Amount, Tax Amount and Unit Cost from Asset Purchase. The second row shows a scatter chart of Original Cost against Current Book Value from Asset Retirement Disposal, a Warranty Status panel that counts Active, Expired, Renewed and Expiring Soon warranties, and a Quick Links panel that

opens the Warranty, Asset Purchase, Maintenance Request and Purchase Items forms. A third row plots Unit Cost against Total for Purchase Items, and the page ends with the embedded All Warranties report.

5.5 Deluge Scripting Implementation

Table 5.3 lists the five Deluge scripts that implement the business logic. Code 5.1 and Code 5.2 show two of the form-event scripts; the complete source of all five scripts is given in Appendix C.

Table 5.3: Deluge Scripts Implemented

Script ID	Trigger / Event	Purpose	Deluge Concepts Used
DS-01	asset_registration → on add / on success	E-mail the assigned employee when an asset is registered	fetch with criteria, isNull check, sendmail
DS-02	maintenance_request → on add or edit / on validate	Block a Critical request that has no technician	if, && operator, isNull, field-level alert, cancel submit
DS-03	asset_transfer → on add / on success	Initialise approval status as Pending	input field assignment
DS-04	maintenance_request → on edit / on validate	Set completion date when repair status is Completed	if, zoho.currentdate
DS-05	Schedule: Daily (start 31-Oct-2026)	Recalculate warranty status for every warranty	for each, date comparison, addDay(30), field update

Code 5.1: Critical Maintenance Priority Validation (DS-02)

```
// Script ID : DS-02
// Location  : maintenance_request -> on add or edit -> on validate
// Purpose   : A Critical request must have a technician assigned
if(input.priority == "Critical" && input.assigned_technician.isNull())
{
    alert(assigned_technician,"A technician must be assigned for critical
priority issues.");
    cancel submit;
}
```

Code 5.2: Asset Assignment Notification (DS-01)

```
// Script ID : DS-01
// Location  : asset_registration -> on add -> on success
// Purpose   : Notify the assigned employee by e-mail
employeeRecord = employee_user[employee_name.first_name ==
input.assigned_employee];
if(!employeeRecord.email.isNull())
{
```

```
    sendmail
    [
        from :zoho.loginuserid
        to :employeeRecord.email
        subject :"Asset Assigned to You"
        message :"Dear " + employeeRecord.employee_name.first_name +
                 ",\n\nYou have been assigned a new asset. Please check the
asset details in the system." +
                 "\n\nAsset Name: " + input.asset_name + "\nAsset Category: " +
input.asset_category +
                 "\n\nThank you!"
    ]
}
```

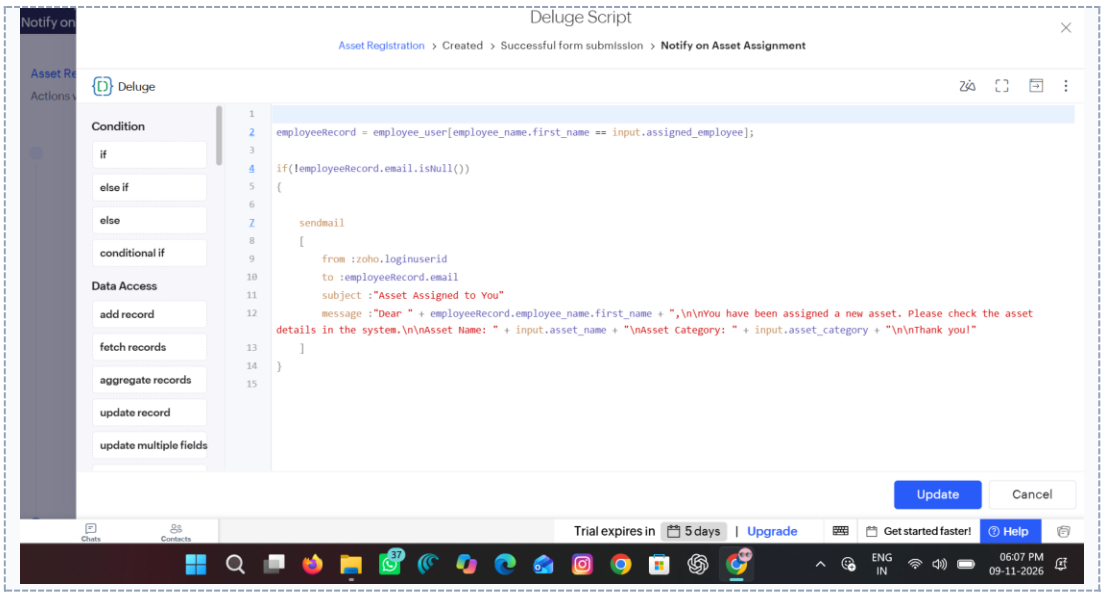


Figure 5.6: Deluge Script Editor and Workflow Configuration

5.6 Scheduled Background Engines

The Warranty Expiry Monitoring workflow runs daily. For each warranty that has an end date and is not marked Renewed, it sets the status to Expired when the end date is in the past, to Expiring Soon when it falls within the next 30 days, and to Active otherwise. This keeps the dashboard counters and warranty reports current without manual updates.

Code 5.3: Daily Warranty Expiry Monitoring (DS-05)

```
// Script ID : DS-05
// Location  : Schedule: Daily (start 31-Oct-2026 12:00:00)
// Purpose   : Refresh the status of every warranty
for each  rec in warranty[ID != 0]
{
    if(rec.end_date != null && rec.warranty_status != "Renewed")
    {
        if(rec.end_date < zoho.currentdate)
```

```
        {
            rec.warranty_status = "Expired";
        }
        else if(rec.end_date <= zoho.currentdate.addDay(30))
        {
            rec.warranty_status = "Expiring Soon";
        }
        else
        {
            rec.warranty_status = "Active";
        }
    }
}
```

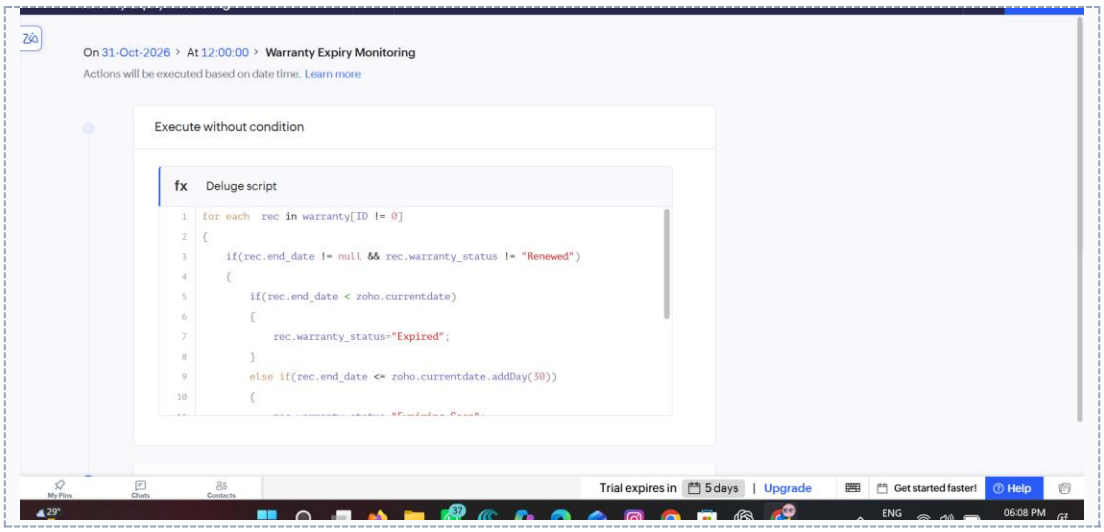


Figure 5.7: Daily Scheduled Workflow Configuration

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Methodology

Testing covered form validations, Deluge event scripts, lookup relationships, the scheduled workflow, report and dashboard output, and profile-based access, using sample vendors, employees, assets and warranties entered through the application's own forms.

6.2 Test Cases & Execution Results

Table 6.1: Test Cases and Execution Results

TC ID	Module	Test Description & Input	Expected Result	Actual Result	Status
TC-01	department	Add a department with name, head, location, e-mail and status Active	Record saved	Saved and listed in All Departments	Pass
TC-02	vendor_supplier	Add a vendor with GST number and category	Vendor Id auto-generated	Vendor Id assigned automatically	Pass
TC-03	asset_purchase	Add a purchase choosing a vendor from the lookup	Saved; appears on Payment Status board	Saved; card shown under its status	Pass
TC-04	asset_registration	Register an asset and leave Asset Status untouched	Asset Id generated; status In Stock	Asset Id generated; status In Stock	Pass
TC-05	DS-01 notification	Register an asset assigned to an existing employee	Assignment e-mail sent to employee	E-mail delivered	Pass
TC-06	DS-02 validation	Save a Critical request without a technician	Blocked with message under the field	Message shown; save cancelled	Pass
TC-07	DS-02 validation	Save a Critical request with a technician	Saved	Saved	Pass
TC-08	DS-04 completion	Edit a request and set Repair Status to Completed	Completion Date set to today	Date populated automatically	Pass
TC-09	DS-03 transfer	Add an asset transfer request	Approval Status = Pending	Status shown as Pending	Pass
TC-10	DS-05 schedule	Run with warranties ended, ending within 30 days, later and Renewed	Expired, Expiring Soon, Active, unchanged	Statuses updated accordingly	Pass
TC-11	RBAC	Employee User opens Asset Registration	Module not available	Module not visible	Pass

TC ID	Module	Test Description & Input	Expected Result	Actual Result	Status
TC-12	Dashboard	Add purchases and check KPI cards	Totals match report aggregates	Totals matched	Pass

6.3 Defect Log & Resolutions

Table 6.2: Defect Log and Design Observations

Bug ID	Description	Root Cause	Fix Applied / Action	Status
BUG-01	A Critical maintenance request could be saved with no technician	No rule linked priority to technician	On-validate script DS-02 shows a field-level error and cancels the save	Closed
BUG-02	Completion date left blank when a repair was finished	Date depended on manual entry	DS-04 stamps the current date when status is Completed	Closed
BUG-03	Warranty status went stale as dates passed	Status was entered once at data entry	Daily schedule DS-05 recalculates status for every warranty	Closed
BUG-04	New transfer requests had no initial approval state	Approval Status left blank on creation	DS-03 sets approval status to Pending on add	Closed
OBS-05	Application and schedule use the America/Los_Angeles time zone	Default workspace time zone	Align both to Asia/Kolkata before institutional use	Open
OBS-06	Vendor, department and assignee are text in Asset Registration	Chosen for quick entry; notification matches on first name	Convert to lookup fields in a future release	Open

6.4 User Acceptance Testing

Table 6.3: User Acceptance Testing Feedback

User / Evaluator	Role Tested	User Feedback	Action Taken
[Evaluator name]	Asset Administrator	[Enter feedback after the UAT session]	[Enter action]
[Evaluator name]	Procurement Officer	[Enter feedback after the UAT session]	[Enter action]
[Evaluator name]	IT Maintenance Technician	[Enter feedback after the UAT session]	[Enter action]
[Evaluator name]	Department Head	[Enter feedback after the UAT session]	[Enter action]

CHAPTER 7

DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Deployment Strategy

The application was built and tested in the development workspace with sample data and then deployed to production on 09-November-2026. The daily Warranty Expiry Monitoring workflow was scheduled from 31-October-2026.

Table 7.1: Deployment Checklist

S.No.	Deployment Checklist Item	Status
1	Forms, reports, dashboard and navigation sections configured	Done
2	All 12 test cases executed and results recorded	Done
3	Permission profiles (six custom profiles) and CEO role configured	Done
4	Daily Warranty Expiry Monitoring schedule enabled	Done
5	Application published and access URL verified	Done

7.2 Performance Optimization

- Lookup-Based References: Child records point to their parent asset, vendor or purchase by ID instead of duplicating its data.
- Platform-Computed Aggregates: Totals, averages, minima and maxima in reports and sums and counts on the dashboard are computed by the platform, not by scripts.
- Single Daily Batch: Warranty status is refreshed once a day, skipping warranties marked Renewed or lacking an end date, instead of on every page load.
- Criteria-Driven Counters: Each Warranty Status counter uses its own criteria (for example warranty_status == "Expired") for fast retrieval.

7.3 Security Controls and Governance

Table 7.2: Security Controls Implemented

Security Control	Implementation Mechanism	Threat Mitigated
Profile-Based Access	Six custom profiles plus the Administrator profile grant only the modules and reports each job needs.	Unauthorised viewing or modification of records.

Security Control	Implementation Mechanism	Threat Mitigated
Portal Isolation	Vendor Representative and Employee User are portal profiles limited to a few modules.	Exposure of internal asset data to external users.
Role Hierarchy	A CEO role sits above subordinate users for oversight.	Lack of managerial visibility.
Validation Hook	DS-02 cancels a Critical request that has no technician.	Unowned high-severity faults.
Transport Encryption	Access to the application is served over HTTPS (TLS) by Zoho Creator.	Eavesdropping and data interception.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Operational Scenario Walkthrough

The lifecycle is demonstrated through a three-step, multi-role walkthrough:

Step 1: Purchase, Registration and Assignment. The Procurement Officer adds a vendor and records an asset purchase with its line items. The Asset Administrator then registers the delivered laptop, assigns it to an employee and saves it. The system generates the Asset Id, sets the status to In Stock, and e-mails the employee through DS-01.

Step 2: Fault Reporting and Maintenance. The employee raises a maintenance request with priority Critical but without a technician. DS-02 blocks the save and displays a message under the Assigned Technician field. After a technician is entered, the request is saved. When the IT Maintenance Technician later sets the Repair Status to Completed, DS-04 fills in the Completion Date automatically.

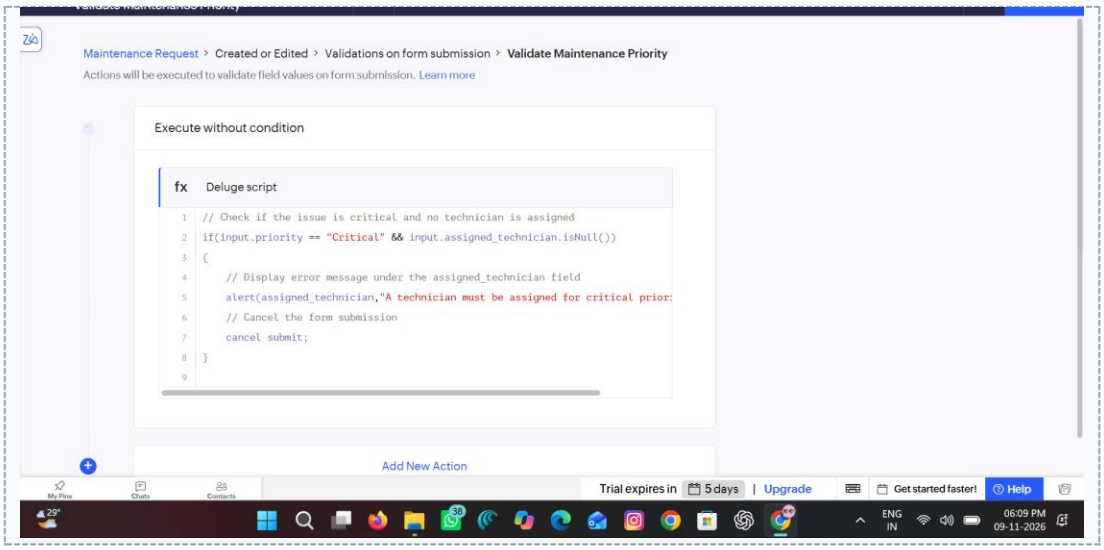


Figure 8.2: Scenario Step 2 – Maintenance Priority Validation

Step 3: Warranty Monitoring and Dashboard. On its next daily run, DS-05 marks warranties as Expired, Expiring Soon or Active according to their end dates. The Warranty Status counters and the cost KPIs on the dashboard then show the current position without any manual compilation.

8.2 Comparison with Existing Manual Systems

Table 8.1: Comparison of Existing Manual System and Proposed Application

Operational Parameter	Existing Manual / Excel Process	Proposed Low-Code Application	Qualitative Gain
Asset Records	Separate files for purchase, assets, warranty and repairs	Eleven linked forms with lookups and auto-numbered identifiers	Single source of truth
Assignment Awareness	Employee informed verbally, if at all	Automatic e-mail on registration	Clear accountability
Warranty Monitoring	Expiry noticed after a failure	Daily schedule sets Active, Expiring Soon, Expired	Timely renewal decisions
Maintenance Control	Informal reports; no technician guarantee	Critical requests need a technician; completion date automatic	Faster, owned repairs
Transfers and Audits	Phone requests; yearly paper check	Structured transfer and audit forms with approval and result fields	Traceable movements
Management Reporting	Manual compilation from several files	Live dashboard and 20 reports with totals	Continuous visibility

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Asset Lifecycle Management System addresses the fragmentation, weak accountability and limited visibility that characterise manual asset tracking. Using Zoho Creator's forms, lookups, reports, ZML dashboard, Deluge scripting and permission profiles, the project covers the complete journey of an asset from purchase to disposal. Assignment notifications, critical-priority validation, automatic completion dates, pending transfer approvals and a daily warranty status engine reduce manual effort and prevent common omissions, while a dashboard and twenty reports give management a live view of cost, status and condition.

9.2 Future Enhancements

- Computed Purchase Values: Calculate tax amount, subtotal and total automatically from quantity, unit cost and tax percentage, and roll purchase items up into the parent purchase.
- Normalised Asset Master: Replace the text fields for vendor, department and assigned employee with lookups to the corresponding master forms.
- Automatic Status Synchronisation: Update Asset Status when a transfer is approved, a repair starts or completes, or a disposal is approved.
- Warranty and Payment Reminders: E-mail owners before warranties expire and when payments remain pending.
- Depreciation and Book Value: Calculate current book value automatically from purchase cost, date and category.
- Mobile Audit with QR Codes: Scan asset labels with the Zoho Creator mobile app to complete audits on site.
- Approval Workflows: Route purchase, transfer and disposal approvals through Zoho Creator approvals with e-mail actions.
- Localisation: Switch the time zone to Asia/Kolkata and the currency to INR for institutional deployment.

9.3 Learning Outcomes and Career Relevance

Developing this project gave practical experience in relational data modelling on a cloud platform, event-driven and scheduled Deluge scripting, dashboard and report design, and profile-based access control. These skills align with roles such as Low-Code Solutions Consultant, Business Process Automation Engineer, Cloud Application Developer and IT Asset Management Analyst.

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APPENDIX A

ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Zoho Creator Feature Coverage Checklist

Unit	Feature Description	Implemented	Where in Application	Report Section	Evidence Ref.
I	Application created & navigation organised	Yes	Workspace with 12 form sections	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	Yes	13 types across 11 forms	4.3	Tables 4.1–4.3
II	Relational lookups between forms	Yes	7 lookup fields	4.2	Fig. 4.2
II	Field validations & event rules	Yes	Critical-priority validation (DS-02)	5.5	Code 5.1
II	Filtered / formatted list reports	Yes	11 list reports with aggregates and colour rules	5.3	Table 5.2
II	Kanban / status reports	Yes	9 Kanban boards	5.3	Table 5.2
II	Summary matrix reports	No	Not implemented	—	—
II	Interactive ZML dashboard	Yes	DashBoard page	5.4	Fig. 5.5
II	Deluge form event workflows	Yes	On add / on success, on validate (4 scripts)	5.5	Codes 5.1–5.2
II	Scheduled background workflow	Yes	Warranty Expiry Monitoring	5.6	Code 5.3
II	Blueprint stage model	Yes	14 stages on Asset Purchase	4.4	Fig. 4.3
II	Automated e-mail notification	Yes	Asset-assignment e-mail (DS-01)	5.5	Code 5.2
III	Role-Based Access Control	Yes	7 profiles and CEO role	4.6	Table 4.4
III	Analytics visualisations	Yes	2 scatter charts and KPI counters	5.4	Fig. 5.5
IV	Performance & governance controls	Yes	Criteria counters, daily batch, profiles	7.2, 7.3	Table 7.2

APPENDIX B

COURSE OUTCOME (CO) MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Addressed in Report	Attainment Level
CO1	Explain the principles, architecture, and business value of low-code development using Zoho Creator.	Chapters 1, 2	High
CO2	Develop business applications using forms, reports, dashboards, and Deluge scripting.	Chapters 4, 5, 6	High
CO3	Implement role-based security, access control, and notification integrations in cloud applications.	Chapters 4, 5, 7	High
CO4	Deploy, secure, and optimize applications following governance and performance best practices.	Chapter 7	High
CO5	Evaluate low-code development strategies, platform limitations, and professional developer pathways.	Chapters 8, 9	High

APPENDIX C

COMPLETE DELUGE SOURCE CODE

C.1 Asset Registration – Notify Assigned Employee

```
// Location : asset_registration -> on add -> on success
// Fetch the assigned employee record using the lookup field
employeeRecord = employee_user[employee_name.first_name ==
input.assigned_employee];
// Check if employee record exists and has a valid email
if(!employeeRecord.email.isNull())
{
    // Send email to the assigned employee
    sendmail
    [
        from :zoho.loginuserid
        to :employeeRecord.email
        subject :"Asset Assigned to You"
        message :"Dear " + employeeRecord.employee_name.first_name +
                 ",\n\nYou have been assigned a new asset. Please check the
asset details in the system." +
                 "\n\nAsset Name: " + input.asset_name + "\nAsset Category: " +
input.asset_category +
                 "\n\nThank you!"
    ]
}
```

C.2 Maintenance Request – Validate Maintenance Priority

```
// Location : maintenance_request -> on add or edit -> on validate
// Check if the issue is critical and no technician is assigned
if(input.priority == "Critical" && input.assigned_technician.isNull())
{
    // Display error message under the assigned_technician field
    alert(assigned_technician,"A technician must be assigned for critical
priority issues.");
    // Cancel the form submission
    cancel submit;
}
```

C.3 Asset Transfer – Initialize Approval Status

```
// Location : asset_transfer -> on add -> on success
input.approval_status = "Pending";
```

C.4 Maintenance Request – Auto Update Completion Date

```
// Location : maintenance_request -> on edit -> on validate
if(input.repair_status == "Completed")
{
```

```
    input.completion_date = zoho.currentdate;
}
```

C.5 Scheduled Engine – Warranty Expiry Monitoring

```
// Location : Schedule: Daily (start 31-Oct-2026 12:00:00)
for each  rec in warranty[ID != 0]
{
    if(rec.end_date != null && rec.warranty_status != "Renewed")
    {
        if(rec.end_date < zoho.currentdate)
        {
            rec.warranty_status = "Expired";
        }
        else if(rec.end_date <= zoho.currentdate.addDay(30))
        {
            rec.warranty_status = "Expiring Soon";
        }
        else
        {
            rec.warranty_status = "Active";
        }
    }
}
```

APPENDIX D
USER MANUAL

D.1 Accessing the Application

1. Open Google Chrome or Microsoft Edge and go to the application URL: https://creatorapp.zoho.in/mohanprasath.p.2024.cse_r285/asset-lifecycle-management-system.
2. Sign in with the credentials assigned to your profile. The left menu then shows the sections available to you, such as Dashboard, Warranty, Asset Purchase, Maintenance Request and Asset Registration.

D.2 Instructions for Procurement Officer

1. Vendors: Open Vendor Supplier and click + Add. Enter the vendor name, contact person, phone, e-mail, address, GST number, category and status.
2. Purchases: Open Asset Purchase and click + Add. Select the vendor, enter the purchase order and invoice numbers, category, quantity, unit cost, tax and payment details, and submit. Add each line item in Purchase Items, selecting the parent purchase.

D.3 Instructions for Asset Administrator

1. Registering Assets: Open Asset Registration and click + Add. Enter the asset details, assigned employee (first name), location and warranty dates. The Asset Id is generated automatically and the status starts as In Stock.
2. Lifecycle Records: Use Asset Transfer, Asset Audit and Asset Retirement Disposal to record movements, physical verification and disposals by selecting the asset from the lookup.

D.4 Instructions for Employees, Technicians and Department Heads

1. Reporting a Fault (Employee): Open Maintenance Request and click + Add. Select the asset, issue type, priority and description. A Critical request cannot be saved until a technician is entered.

2. Handling a Request (IT Technician): Open the request, update the repair status and costs, and set the status to Completed when finished; the completion date is filled in automatically.
3. Transfers (Department Head): Open Asset Transfer, add the new employee, department and location with a reason; the approval status starts as Pending.

APPENDIX E

EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	SANDEEP K	Register No.	2116240701464
Project Title	Asset Lifecycle Management System	Date	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups, report types (Kanban, Matrix), ZML UI dashboards	15	
4	Deluge Scripting & Workflows	Depth of scripts; form validations, cascading updates, scheduled engines	15	
5	Integration, RBAC & Security	Automated e-mail notifications, role-based permission profiles, transport security	10	
6	Testing, Deployment & Governance	Test coverage, defect logs, query optimisation, security controls, audit logs	10	
7	Live Demonstration	End-to-end working of application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, code, references	5	
9	Viva-Voce	Understanding of low-code concepts, Deluge syntax, design decisions	10	
	TOTAL		100	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____